

Gustavo Vargas Hakim

MACHINE LEARNING RESEARCH SCIENTIST | MULTIMODAL REPRESENTATION LEARNING | COMPUTER VISION

Montreal, QC

☎ (+1) 438 522 3239 | ✉ vargashakimg@gmail.com | 🌐 <https://gustavovargashakim.github.io/>
📄 github.com/GustavoVargasHakim | 🔗 [linkedin.com/in/gustavoavh](https://www.linkedin.com/in/gustavoavh)

SUMMARY

Machine learning research scientist with a PhD in test-time adaptation and industry R&D experience in multimodal video understanding. Early contributor to **test-time adaptation of vision-language models**, with 7 papers at **NeurIPS, CVPR, ICCV and WACV** (6 as co-first author). At Coactive AI, turned research ideas into shipped methods: **RL finetuning of LLMs (GRPO)**, embedding calibration and **multimodal fusion**. Interested in representation learning, robustness to distribution shift and multimodal models; currently exploring self-supervised learning, world models and energy-based models.

EXPERIENCE

Machine Learning Research Scientist

05/2026 – 09/2026

Coactive AI

Montreal, Canada

- Led **multimodal late fusion** (vision, text, audio) for few-shot video tagging, learning per-tag and per-shot fusion weights; improved AP by **28% relative** over the production baseline
- Designed a modular **active-sampling framework** for few-shot tag learning, combining automatic thresholding (Otsu, elbow, z-score) with weighting functions
- Developed a multimodal **video segmentation** method grouping shots into coherent scenes, with a VLM-annotated evaluation framework

Machine Learning Research Developer

07/2025 – 05/2026

Coactive AI

Montreal, Canada

- Finetuned Qwen models with **GRPO** and **LoRA** to write CLIP-style prompts using a VLM as reward; beat GPT-5.2 by up to **6.3 F1 points** on FineVideo (43K videos), replacing a paid API
- Built an embedding-level **calibration framework** (Platt scaling, hierarchy-aware negative mining) for Amazon Nova across four modalities; shipped to production
- Led audio R&D: music detection and mood recognition, a speech-emotion benchmark with a new **hallucination metric** for audio-language models, and a 5.6× faster loading pipeline

PhD Researcher, Machine Learning

09/2021 – 07/2025

LIVIA, ÉTS Montréal

Montreal, Canada

- Among the early works on **test-time training**, using normalizing flows, mutual-information clustering and noise-contrastive estimation
- Extended test-time adaptation to **vision-language models** (CLIP) before the area was established, and to histopathology VLMs
- Research visit at **Sorbonne Université** with Prof. Nicolas Thome, leading to CLIPTTA (NeurIPS 2025)

Computer Vision Research Intern (Industrial Research Chair)

01/2022 – MM/2025

Matrox Imaging (now Zebra Technologies)

Montreal, Canada

- Researched and benchmarked **industrial anomaly detection** methods on MVTec AD and clients' inspection data
- Adapted models to real production data with the engineering team, and sped up CPU inference by 2.6×

SELECTED PUBLICATIONS

CLIPTTA: Robust Contrastive Vision-Language Test-Time Adaptation

NeurIPS 2025 | Lafon, M.* , **Vargas Hakim, G. A.***, Rambour, C., Desrosiers, C., Thome, N.

Histopath-C: Towards Realistic Domain Shifts for Histopathology Vision-Language Adaptation

WACV 2026 | Noori, M.* , **Vargas Hakim, G. A.***, Osowiechi, D., et al.

CLIPArTT: Adaptation of CLIP to New Domains at Test Time

WACV 2025 | **Vargas Hakim, G. A.***, Osowiechi, D.* , et al.

WATT: Weight Average Test-Time Adaptation of CLIP

NeurIPS 2024 | Osowiechi, D.* , Noori, M.* , **Vargas Hakim, G. A.**, et al.

NC-TTT: A Noise Contrastive Approach for Test-Time Training

CVPR 2024 | Osowiechi, D.* , **Vargas Hakim, G. A.***, et al.

ClusT3: Information Invariant Test-Time Training

ICCV 2023 | **Vargas Hakim, G. A.***, Osowiechi, D.* , et al.

TTTFlow: Unsupervised Test-Time Training With Normalizing Flow

WACV 2023 | Osowiechi, D.* , **Vargas Hakim, G. A.***, et al.

Hybrid Encodings for Neuroevolution of Convolutional Neural Networks: A Case Study

GECCO 2021 | **Vargas Hakim, G. A.**, Mezura-Montes, E., Acosta-Mesa, H.-G.

EDUCATION

PhD in Machine Learning

École de Technologie Supérieure (ÉTS)

09/2021 – 07/2025

Montreal, Canada

- Thesis: Test-Time Adaptation of Computer Vision Models. Advisors: Prof. Christian Desrosiers and Prof. Ismail Ben Ayed

MSc in Artificial Intelligence

Universidad Veracruzana

08/2019 – 07/2021

Xalapa, Mexico

- Thesis: Neuroevolution of Convolutional Neural Networks

BEng in Mechatronics Engineering

UPAEP

08/2013 – 01/2019

Puebla, Mexico

- Exchange semester at Oklahoma State University (Stillwater, US)

SKILLS

Research areas: Representation learning, vision-language and multimodal models, test-time adaptation and distribution shift, self-supervised and contrastive learning, LLM finetuning (GRPO, LoRA), calibration and probabilistic fusion, video and audio understanding, anomaly detection

Models: CLIP, Perception Encoder, SAM3, Qwen3 / Qwen3-VL / Qwen3-Omni, CLAP, AST, Wav2Vec2

Frameworks: Python, PyTorch, Hugging Face (Transformers, TRL, PEFT), scikit-learn, NumPy, pandas, SciPy, OpenCV, FAISS, Anomalib

Compute & tools: HPC clusters, AWS (EC2, S3), Ray, Comet, Git, Claude Code, ~~LaTeX~~ $\text{\LaTeX}$, Notion, Figma

Languages: Spanish (native), English (fluent), French (fluent; TCF Canada, B2–C1)

TEACHING & KNOWLEDGE SHARING

Research knowledge sharing

Coactive AI

2025 – 2026

Montreal, Canada

- Presented paper reviews (NeurIPS 2025) and internal tutorials on RLHF, DPO, GRPO and parameter-efficient finetuning to the research team

Tutorial Lecturer

Summer School on Deep Learning and Medical Imaging

07/2022

Montreal, Canada

- Taught semi-supervised medical image segmentation

Lecturer

Universidad Euro Hispanoamericana

02/2019 – 06/2019

Xalapa, Mexico

- Taught linear algebra, control theory and electromechanical systems